Chapter 6 Electromagnetic Waves at Boundaries

In the previous chapter, we explored how Maxwell's equations give rise to electromagnetic waves in free space and homogeneous media. We derived general solutions such as plane waves and spherical waves, examined energy transport via the Poynting vector, and analyzed the radiation patterns from elementary sources.

In this chapter, we shift our attention to how electromagnetic waves behave when they encounter material boundaries or are confined by structures. Specifically, we will study the phenomena of reflection and transmission at planar interfaces, and then extend our analysis to guided wave propagation in waveguides.

These topics form the theoretical foundation of modern optics, photonics, and microwave engineering. Whether in fiber-optic communication, laser systems, or RF signal transport, understanding how EM waves reflect, refract, and propagate within bounded media is essential.

1. Transmission and Reflection at a Planar Interface

In this section, we study the transmission and reflection of a plane electromagnetic wave at the boundary between two linear, isotropic, and homogeneous media. The media are assumed to be non-magnetic (i.e., relative permeability $\mu_{\mathrm{r}} = 1$) and characterized by a refractive index $n \geq 1$, where n is a real number (which may be generalized to complex values to account for absorption or dispersion).

1) Introduction and Preparatory Concepts

① Preparation 1: Wave Impedance

For a plane wave in a source-free, homogeneous medium, the electric field $\vec{E}$, magnetic field $\vec{B}$, and wave vector $\vec{k}$ are mutually orthogonal:

$$\vec{E} \perp \vec{B} \perp \vec{k}, \quad |\vec{B}| = \frac{|\vec{E}|}{v},$$

where v is the speed of light in the medium, given by:

$$v = \frac{1}{\sqrt{\varepsilon\mu}} = \frac{c}{n}.$$

The magnetic field intensity $\vec{H}$ is related to $\vec{B}$ via $\vec{B} = \mu\vec{H}$, hence:

$$|\vec{H}| = \frac{|\vec{B}|}{\mu} = \frac{|\vec{E}|}{\mu v}.$$

This leads to the ratio of electric to magnetic field magnitudes, defined as the wave impedance:

$$\eta = \frac{|\vec{E}|}{|\vec{H}|} = \sqrt{\frac{\mu}{\varepsilon}}.$$

For free space, this becomes:

$$\eta_0 = \sqrt{\frac{\mu_0}{\varepsilon_0}} \approx 377\,\Omega.$$

For two non-magnetic ($\mu_r = 1$) and non-absorbing media, the ratio of their impedances is:

$$\frac{\eta_1}{\eta_2} = \sqrt{\frac{\varepsilon_2}{\varepsilon_1}} = \frac{n_2}{n_1}.$$

② **Preparation 2: Coherent Wave Interference**

Two coherent waves—that is, waves with a fixed phase relationship—can interfere. In this course, we will assume all interfering waves are coherent. When two coherent waves propagate in opposite directions, their superposition leads to the formation of a standing wave, characterized by nodes and antinodes of oscillating electric and magnetic fields.

Example 1: Standing wave formed by counter-propagating plane waves

Consider two counter-propagating waves with the same amplitude and frequency:

$$\tilde{u}_1 = \tilde{A}e^{j(kz-\omega t)}, \quad \tilde{u}_2 = \tilde{A}e^{-j(kz+\omega t)}.$$

By the principle of superposition, the total field is:

$$\tilde{u} = \tilde{u}_1 + \tilde{u}_2 = \tilde{A}\left(e^{jkz} + e^{-jkz}\right)e^{-j\omega t} = 2\tilde{A}\cos(kz)\ e^{-j\omega t}.$$

Taking the real part, which corresponds to the physical (measurable) field:

$$\mathrm{Re}[\tilde{u}] = 2A\cos(kz)\cos(\omega t - \phi),$$

where ϕ is the phase of $\tilde{A}$. We see that the result is a standing wave, with spatial oscillation $\cos(kz)$ and temporal oscillation $\cos(\omega t)$.

Example 2: Partial standing wave from two beams at an angle

Now consider two obliquely propagating waves:

$$\tilde{u}_1 = A_1 e^{j(k_x x + k_z z - \omega t)}, \quad \tilde{u}_2 = A_2 e^{j(-k_x x + k_z z - \omega t)},$$

where A_1 and A_2 are real amplitudes and $A_1 > A_2$. The superposition gives:

$$\tilde{u} = A_1 e^{j(k_x x + k_z z - \omega t)} + A_2 e^{j(-k_x x + k_z z - \omega t)}$$

$$= (A_1 - A_2)e^{j(k_x x + k_z z - \omega t)} + A_2\left(e^{jk_x x} + e^{-jk_x x}\right)e^{j(k_z z - \omega t)}$$

$$= \Delta A e^{j(\vec{k}\cdot\vec{r} - \omega t)} + 2A_2 \cos(k_x x)\, e^{j(k_z z - \omega t)},$$

where $\Delta A = A_1 - A_2$, and $\vec{k} = (k_x, k_z)$. This expression can be interpreted as:

- A plane wave propagating along $\vec{k}$, with reduced amplitude ΔA,
- Superimposed with a partially standing wave pattern in the x-direction due to the $\cos(k_x x)$ term, modulated along z.

This illustrates how unequal-amplitude beams propagating at different directions can lead to a mixture of traveling and standing wave components.

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2) Snell's Law

① Wave vectors and geometry

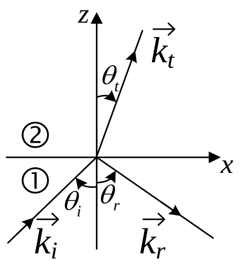

Figure. Geometry of wave reflection and transmission.

Let:

- $\vec{k_i}, \vec{k_r}$, and $\vec{k_\mathrm{t}}$ denote the wave vectors of the incident, reflected, and transmitted waves, respectively.
- The interface lies in the x–y plane (i.e., z = 0), and is uniform along those directions.
- The incident wave propagates in the x–z plane, called the plane of incidence.
- We fix the **y**-axis to be perpendicular to this plane.

Thus, the incident wave vector has components:

$$\vec{k_i} = k_{ix}\hat{x} + k_{iz}\hat{z}, \text{ with } \vec{k_i}\cdot\vec{r} = k_{ix}x + k_{iz}z.$$

The incident electric field (in complex notation) is:

$$\tilde{\vec{E}}_i(\vec{r}, t) = \tilde{\vec{E}}_{0i}\, e^{j(k_{ix}x + k_{iz}z - \omega t)},$$

where $\tilde{\vec{E}}_{0i}$ is the complex amplitude.

② Boundary matching conditions

To apply the boundary conditions, we write the electric fields on both sides of the interface z = 0:

- In medium ① (incident + reflected waves):

$$\vec{\tilde{E}}_1(x, z = 0, t) = \vec{\tilde{E}}_{0i}\, e^{i(k_{ix}x - \omega t)} + \vec{\tilde{E}}_{0r} e^{i(k_{rx}x - \omega t)}.$$

- In medium ② (transmitted wave only):

$$\vec{\tilde{E}}_2(x, z = 0, t) = \vec{\tilde{E}}_{0t}\, e^{i(k_{tx}x - \omega t)}.$$

We use $\tilde{E}_0$ to denote the electric field's complex amplitude, and use "i", "r", "t" for the incident, reflected, and transmitted waves, respectively. To ensure that the tangential components of the electric and magnetic fields are continuous across the boundary z = 0, the exponential factors of all waves must match in both space and time. This requires:

$$\omega_i = \omega_r = \omega_t, \quad k_{ix} = k_{rx} = k_{tx}.$$

Using $k = n\omega/c = nk_0$, we can express these components as:

$$k_{ix} = k_i \sin\theta_i = n_1 k_0 \sin\theta_\mathrm{i},$$

$$k_{rx} = k_r \sin\theta_r = n_1 k_0 \sin\theta_r,$$

$$k_{tx} = k_t \sin\theta_t = n_2 k_0 \sin\theta_t.$$

Since $k_{ix} = k_{rx} = k_{tx}$, it follows that:

$$\boxed{\theta_r = \theta_i} \quad \text{(Law of Reflection)},$$

$$\boxed{n_1 \sin\theta_i = n_2 \sin\theta_t} \quad \text{(Snell's Law)}.$$

This fundamental result governs how light bends at a dielectric interface.

3) Fresnel equations

The Fresnel equations describe how light (or more generally, electromagnetic waves) behaves when it encounters an interface between two media with different refractive indices. Specifically, they quantify how much of the wave is reflected and how much is transmitted (refracted) at the boundary. Since any electromagnetic field can be decomposed into two orthogonally polarized components—one with the electric field perpendicular to the plane of incidence (S-wave, where "S" stands for *senkrecht*, German for "perpendicular"), and the other with the electric field parallel to the plane of incidence (P-wave, where "P" stands for *parallel*)—each polarization satisfies a different set of boundary conditions. Therefore, we analyze them separately.

① S-Polarization (Electric Field Perpendicular to the Plane of Incidence):

In the following derivation, we focus on the amplitude reflection coefficient. Other parameters, such as transmission coefficients or intensities, can be derived in a similar manner.

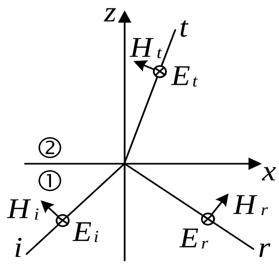

Figure. Field configuration for S-polarized wave.

The incident electric field $\vec{E}_i$ is perpendicular to the plane of incidence (the xz-plane), and thus lies along the $\hat{y}$ direction. The associated magnetic field is determined by the relation $\vec{k} \propto \vec{E} \times \vec{H}$. We apply the boundary conditions at the interface between medium ① and medium ②.

Boundary Condition 1: Continuity of Tangential Electric Fields

We begin by writing out the electric fields on the boundary (at z = 0).

In medium ①, both the incident and reflected waves exist, with a total electric field:

$$(\tilde{E}_{0i} + \tilde{E}_{0r})e^{i(k_x x - \omega t)}.$$

Note that $k_z z = 0$ because z = 0.

In medium ②, only the transmitted wave propagates, thus the electric field is

$$\tilde{E}_{0t}\, e^{i(k_x x - \omega t)}.$$

The continuity of the electric fields requires that

$$\tilde{E}_{0i} + \tilde{E}_{0r} = \tilde{E}_{0t}. \tag{1}$$

Boundary Condition 2: Continuity of Tangential H Fields

The H field components are related to the electric field via:

$$\vec{H} = \frac{1}{\eta}\hat{k} \times \vec{E}.$$

In medium ① (incident + reflected), the magnetic field has a tangential x-component, and we must account for the reflected wave's direction (which flips the sign of the x-component):

$$-\frac{1}{\eta_1}\tilde{E}_{0\mathrm{i}} \cos\theta_i + \frac{1}{\eta_1}\tilde{E}_{0\mathrm{r}} \cos\theta_i = \frac{1}{\eta_2}\tilde{E}_{0\mathrm{t}} \cos\theta_t. \tag{2}$$

(Note: the minus sign appears because the reflected wave propagates in the opposite x-direction.)

Equation (1) can be converted to:

$$\frac{1}{\eta_2}\cos\theta_t\left(\tilde{E}_{0i}+\tilde{E}_{0\mathrm{r}}\right)=\frac{1}{\eta_2}\cos\theta_t\,\tilde{E}_{0t}. \tag{3}$$

Adding equations (2) and (3) yields:

$$\tilde{E}_{0i}\left(\frac{1}{\eta_2}\cos\theta_t-\frac{1}{\eta_1}\cos\theta_i\right)+\tilde{E}_{0\mathrm{r}}\left(\frac{1}{\eta_2}\cos\theta_t+\frac{1}{\eta_1}\cos\theta_i\right)=0$$

Solving for the amplitude reflection coefficient:

$$\tilde{r}_s\equiv\frac{\tilde{E}_{0\mathrm{r}}}{\tilde{E}_{0\mathrm{i}}}=\frac{\eta_2\cos\theta_i-\eta_1\cos\theta_t}{\eta_2\cos\theta_i+\eta_1\cos\theta_t}$$

Using $\frac{\eta_1}{\eta_2}=\frac{n_2}{n_1}$ (intrinsic impedance in non-magnetic media), this becomes:

$$\boxed{\tilde{r}_s=\frac{n_1\cos\theta_i-n_2\cos\theta_t}{n_1\cos\theta_i+n_2\cos\theta_t}}.$$

② P-Polarization (Electric Field in the Plane of Incidence)

We now consider the case of P-polarized light, where the electric field lies in the plane of incidence (the xz-plane), see the Figure below. As before, we aim to derive the amplitude reflection coefficient.

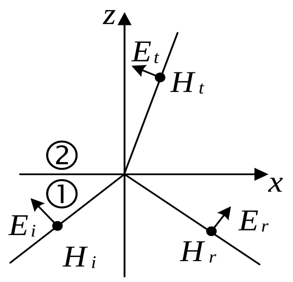

Figure. Field configuration for P-polarized wave.

<u>Boundary Condition 1: Continuity of Tangential Electric Field</u>

The tangential component of the electric field is along the surface (x-direction), so we project $\vec{E}$ onto the x-axis:

$$-\tilde{E}_{0i}\cos\theta_i+\tilde{E}_{0\mathrm{r}}\cos\theta_i=-\tilde{E}_{0\mathrm{t}}\cos\theta_t \tag{4}$$

The minus signs arise from field direction conventions.

<u>Boundary Condition 2: Continuity of Tangential H Field</u>

The tangential magnetic field is proportional to the electric field amplitude:

$$\frac{1}{\eta_1}\tilde{E}_{0i}+\frac{1}{\eta_1}\tilde{E}_{0\mathrm{r}}=\frac{1}{\eta_2}\tilde{E}_{0\mathrm{t}} \tag{5}$$

We manipulate Equations (4) and (5) to eliminate $\tilde{E}_{0\mathrm{t}}$ and solve for $\tilde{r}_p$.

Multiply Eq. (5) by $\cos\theta_t$:

$$\frac{1}{\eta_1}\cos\theta_t\,\tilde{E}_{0i} + \frac{1}{\eta_1}\cos\theta_t\;\tilde{E}_{0\mathrm{r}} = \frac{1}{\eta_2}\cos\theta_t\;\tilde{E}_{0\mathrm{t}} \tag{6}$$

Multiply Eq. (4) by $\frac{1}{\eta_2}$:

$$-\frac{1}{\eta_2}\tilde{E}_{0i}\cos\theta_i + \frac{1}{\eta_2}\tilde{E}_{0\mathrm{r}}\cos\theta_i = -\frac{1}{\eta_2}\tilde{E}_{0\mathrm{t}}\cos\theta_t \tag{7}$$

Add Eqs. (6) and (7):

$$\left(\frac{1}{\eta_1}\cos\theta_t - \frac{1}{\eta_2}\cos\theta_i\right)\tilde{E}_{0i} + \left(\frac{1}{\eta_1}\cos\theta_t + \frac{1}{\eta_2}\cos\theta_i\right)\tilde{E}_{0\mathrm{r}} = 0.$$

Solving for the reflection coefficient:

$$\tilde{r}_p = \frac{\tilde{E}_{0\mathrm{r}}}{\tilde{E}_{0i}} = \frac{\frac{1}{\eta_2}\cos\theta_i - \frac{1}{\eta_1}\cos\theta_t}{\frac{1}{\eta_2}\cos\theta_i + \frac{1}{\eta_1}\cos\theta_t}$$

Using $\frac{\eta_1}{\eta_2} = \frac{n_2}{n_1}$ in non-magnetic media, this becomes:

$$\boxed{\tilde{r}_p = \frac{n_2\cos\theta_i - n_1\cos\theta_t}{n_2\cos\theta_i + n_1\cos\theta_t}}.$$

③ Case 1: $n_1 < n_2$: Light Incident from Air to Glass

Let us consider an electromagnetic wave incident from a medium with refractive index n_1 into a denser medium with $n_2 > n_1$, such as from air ($n_1 = 1$) into glass ($n_2 = 1.5$). According to Snell's law, $\theta_t < \theta_i$, as illustrated in the following figure.

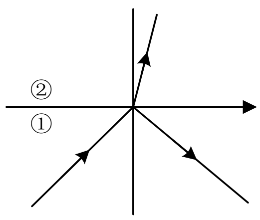

Figure. Ray directions for $n_1 < n_2$.

The amplitude reflection coefficient $\tilde{r}$ depends on both polarization and incidence angle. The table below lists two representative angles ($\theta_i = 0°$ and $90°$):

Polarization	$\theta_i = 0°$	$\theta_i = 90°$
S	$\frac{n_1 - n_2}{n_1 + n_2}$	-1
P	$\frac{n_2 - n_1}{n_2 + n_1}$	-1

Table. Amplitude coefficient at $\theta_i = 0°$ and $90°$ for $n_1 < n_2$.

In the figure below, the amplitude reflection coefficient as a function of incidence angle for $n_1 = 1, n_2 = 1.5$ is plotted.

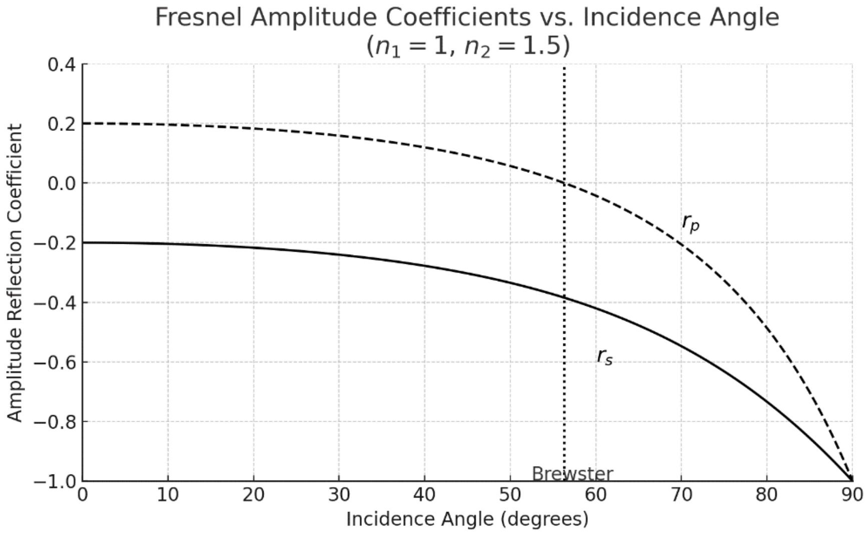

Figure. $\tilde{r}$ as a function of the incidence angle.

For P-polarized light, there exists a specific incidence angle at which the reflection coefficient becomes exactly zero. This special angle is called Brewster's angle, and it satisfies:

$$\tilde{r}_p = \frac{n_2 \cos\theta_i - n_1 \cos\theta_t}{n_2 \cos\theta_i + n_1 \cos\theta_t} = 0$$

This occurs when:

$$n_2 \cos\theta_i = n_1 \cos\theta_t.$$

Using Snell's law:

$$n_1 \sin\theta_i = n_2 \sin\theta_t.$$

We eliminate n_1 and n_2 and derive the condition:

$$\theta_i + \theta_t = 90° \quad \Rightarrow \quad \vec{k}_i \perp \vec{k}_t$$

This means that the reflected and transmitted rays are orthogonal at Brewster's angle.

Solving the above condition yields Brewster's angle:

$$\boxed{\theta_B = \tan^{-1}\left(\frac{n_2}{n_1}\right)}$$

For air-to-glass ($n_1 = 1, n_2 = 1.5$):

$$\theta_B = \tan^{-1}(1.5) \approx 56.3°.$$

At θ_B, the reflected component of P-polarized light vanishes entirely, resulting in a reflected beam that is purely S-polarized. This phenomenon has important practical applications. In optical coatings and polarizers, Brewster's angle is exploited to selectively generate or filter polarized light. In laser systems, Brewster-angle windows are commonly used to minimize reflection losses for P-polarized beams, thereby improving output efficiency. In photography and imaging, polarizing filters utilize this effect to suppress glare and enhance contrast by blocking certain polarized components of reflected light.

Note:

If $n_1 = 1$ and $\tilde{n}_2 = n + j\kappa$ ($\tilde{n}_2$ has an imaginary part), and consider normal incidence $\theta_i = 0$, the reflectance becomes:

$$R = |\tilde{r}|^2 = \left|\frac{\tilde{n}_2 - 1}{\tilde{n}_2 + 1}\right|^2 = \frac{(n-1)^2 + \kappa^2}{(n+1)^2 + \kappa^2}$$

This is valid for both S and P polarizations and reflects absorption loss due to $\kappa \neq 0$.

④ Case 2: Total Internal Reflection: $n_1 > n_2$ (e.g., Water to Air)

Let us consider an electromagnetic wave incident from a medium with refractive index n_1 into a rarer medium with $n_1 > n_2$, such as from glass ($n_1 = 1.5$) into air ($n_2 = 1$). According to Snell's law, $\theta_t > \theta_i$, as illustrated in the following figure.

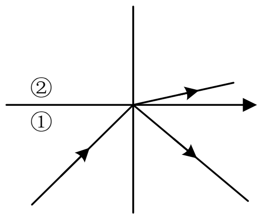

Figure. Ray directions for $n_1 > n_2$.

This case is fundamentally different from the air-to-glass case. When light travels from a denser medium (n_1) to a rarer medium (n_2), the behavior of the transmitted wave changes dramatically as the incidence angle increases.

From Snell's Law:

$$n_1 \sin\theta_i = n_2 \sin\theta_t,$$

when the transmitted angle reaches $\theta_t = 90°$, we obtain the critical angle θ_c:

$$\sin\theta_c = \frac{n_2}{n_1}.$$

For $\theta_i > \theta_c$, the equation predicts $\sin\theta_t > 1$, which is not physically meaningful for real angles. This signals the onset of <u>total internal reflection</u>, where the wave is no longer transmitted in the usual sense.

At $\theta_i = \theta_c$, we have $\cos\theta_t = 0$, which leads to:

$$\tilde{r}_s = \tilde{r}_p = 1.$$

When $\theta_i > \theta_c$, Snell's law no longer yields a real θ_t, we observe that:

$$k_{tx} = n_2 k_0 \sin\theta_t = k_{ix},$$

$$k_{tz}^2 = k_2^2 - k_{tx}^2 < 0,$$

which implies k_{tz} is purely imaginary. At $\theta_i = 90°$, we can verify directly that:

$$\tilde{r}_s = \tilde{r}_p = -1, \quad |\tilde{r}_s| = |\tilde{r}_p| = 1, \quad \angle\tilde{r}_s = \angle\tilde{r}_p = -\pi.$$

For intermediate angles $\theta_c < \theta_i < 90°$, the magnitude remains $|\tilde{r}| = 1$, but the phase shifts $\angle\tilde{r}_s$ and $\angle\tilde{r}_p$ vary within the interval $[0, -\pi]$. The amplitude reflection coefficient $\tilde{r}$ depends on both polarization and incidence angle. The table below lists three representative angles:

Polarization	$\theta_i = 0°$	$\theta_i = \theta_c$	$\theta_i = 90°$
S	$\dfrac{n_1 - n_2}{n_1 + n_2}$	1	$-1 = e^{-j\pi}$
P	$\dfrac{n_2 - n_1}{n_2 + n_1}$	1	$-1 = e^{-j\pi}$

Table. Amplitude coefficient at $\theta_i = 0°$ and $90°$ for $n_1 > n_2$.

In the figure below, the amplitude reflection coefficient as a function of incidence angle for $n_1 = 1.5, n_2 = 1$ is plotted.

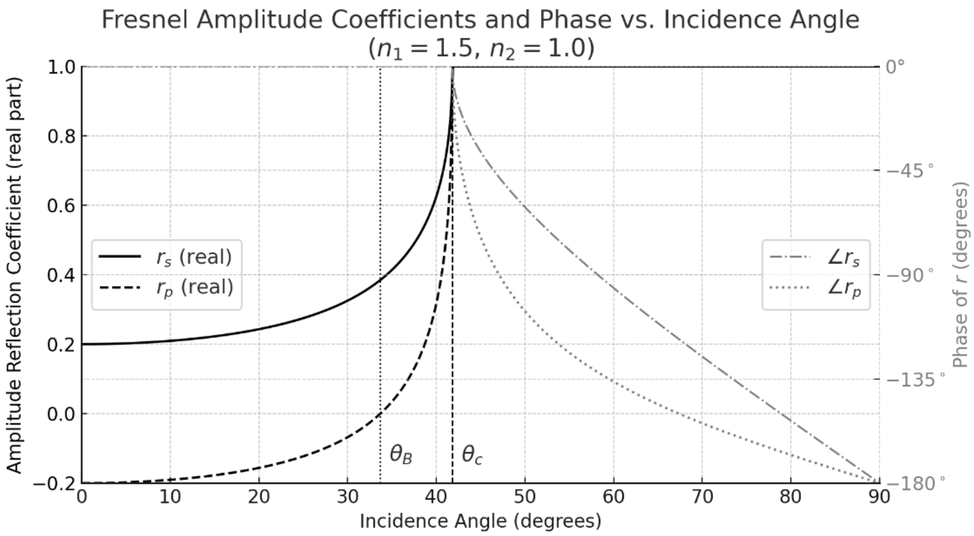

Figure. Amplitude and phase of $\tilde{r}$ as a function of the incidence angle.

The left y-axis shows the real part of the reflection coefficients. For incidence angles below the critical angle θ_c, both $\tilde{r}_s$ and $\tilde{r}_p$ are real-valued, and their amplitudes increase with angle. At Brewster's angle θ_B, the coefficient $\tilde{r}_p$ vanishes, indicating zero reflection for P-polarized light. Beyond the critical angle, the reflection coefficients become complex with unit magnitude, and the plot transitions to show their phase shifts (in degrees) on the right y-axis. Notably, the phases of both $\tilde{r}_s$ and $\tilde{r}_p$ undergo a rapid decrease, approaching $-\pi$ as the angle approaches $90°$. This phase behavior reflects the total internal reflection regime, where the transmitted field becomes evanescent, and energy is entirely reflected with a phase lag. The clear separation and labeling of amplitude and phase characteristics across both axes provide an intuitive view of how reflection transforms at dielectric interfaces.

Notes:

Consider the following field configuration, in which an S-polarized electromagnetic wave is incident from an optically denser medium onto a rarer medium ($n_2 < n_1$). The diagram illustrates the case where the incidence angle θ_i is less than the critical angle θ_c, such that both reflection and transmission occur. While the figure depicts a partial reflection regime, the underlying field relations can be extended to the case of total internal reflection. In that regime, the transmitted wave becomes evanescent.

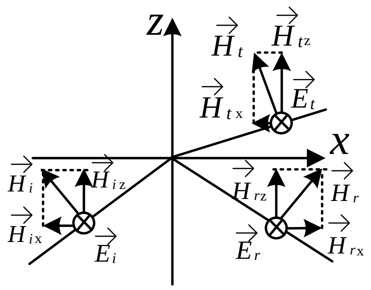

Figure. Field configuration for an S-polarized wave incident from a denser medium.

For $\theta_i < \theta_c$, the reflected and incident waves are in phase. For example, if $\tilde{r} = 0.7$, then:

$$\tilde{E}_{0i} = 1, \quad \tilde{E}_{0r} = 0.7, \quad \tilde{E}_{0t} = 1.7,$$

satisfying the boundary condition:

$$\tilde{E}_{0i} + \tilde{E}_{0r} = \tilde{E}_{0t}.$$

At first glance, the transmitted amplitude may appear "larger" than the incident wave, which raises a question about energy conservation. However, this is resolved by analyzing the Poynting vector, which depends on both field amplitude and wave impedance.

The time-averaged Poynting vector $\langle\vec{S}\rangle \propto nE^2$, and specifically its normal component is:

In medium 1 (incident side):

$$\langle S_z^{(1)}\rangle = n_1 \cos\theta_i \left(\left|\tilde{E}_{0i}\right|^2 - \left|\tilde{E}_{0r}\right|^2\right),$$

whereas in medium 2 (transmitted side):

$$\langle S_z^{(2)}\rangle = n_2 \cos\theta_t \left|\tilde{E}_{0t}\right|^2.$$

Substituting the Fresnel coefficients and simplifying confirms that:

$$\langle S_z^{(1)}\rangle = \langle S_z^{(2)}\rangle,$$

verifying that energy is conserved despite apparent amplitude differences.

Now we analyze what happens when the incident angle increases beyond the critical angle ($\theta_i > \theta_c$). In the rarer medium (medium 2), the transmitted electric field (S-polarization, which oscillates in the y-direction) takes the form:

$$\tilde{E}_t(x,z,t) = \tilde{E}_{0t}\, e^{j(k_x x-\omega t)}\, e^{jk_z z} = \tilde{E}_{0t} e^{-\kappa z}\, e^{j(k_x x-\omega t)},$$

indicating oscillatory behavior along x and exponential decay along z. The corresponding magnetic field components become complex-valued:

$$\tilde{H}_x = -\frac{\tilde{E}_t}{\eta_2}\cos\theta_t = -\frac{\tilde{E}_t}{\eta_2}\frac{j\kappa}{k_2},$$
$$\tilde{H}_z = \frac{\tilde{E}_t}{\eta_2}\sin\theta_t = \frac{\tilde{E}_t}{\eta_2}\frac{k_1\sin\theta_i}{k_2}.$$

and the time-averaged power flow shows:

$$\langle S_z^{(2)}\rangle = Re\left[\frac{1}{2}\tilde{E}_y\left(-\tilde{H}_x^*\right)\right] = \frac{1}{2}Re\left[-\frac{j\kappa}{\eta_2 k_2}\left|\tilde{E}_{0t}\right|^2 e^{-2\kappa z}\right] = 0,$$

which means there is no net energy flows into the rarer medium, consistent with total reflection. For the tangential component

$$\langle S_x^{(2)}\rangle = Re\left[\frac{1}{2}\tilde{E}_y\tilde{H}_z^*\right] = \frac{1}{2}\frac{\left|\tilde{E}_{0t}\right|^2}{\eta_2 k_2}k_1\sin\theta_i\, e^{-2\kappa z},$$

indicating that energy flows tangentially along the interface.

The figure below shows the normalized electric field distribution at $t = 0$ for an S-polarized plane wave undergoing total internal reflection at an interface between a denser medium ($n_1 = 1.5$) and a rarer medium ($n_2 = 1.0$). The wavelength of the wave is 1 μm, the angle of incidence is chosen slightly above the critical angle, resulting in complete reflection and the formation of an evanescent wave in the rarer medium. In the lower half-space ($z < 0$), the incident and reflected waves interfere to form a standing wave pattern. In the upper half-space ($z > 0$), the transmitted wave does not propagate in the z direction but instead decays exponentially away from the interface while maintaining oscillations and propagates along the x-direction. This configuration clearly illustrates the physical mechanism of total internal reflection and the nature of evanescent fields. Although this snapshot is taken at $t = 0$, as time evolves, the entire interference pattern in both regions shifts to the right at a constant phase velocity $v_p = \omega/k_x$. This demonstrates that despite the absence of energy transport into the rarer medium, the field maintains a coherent wave pattern moving along the interface.

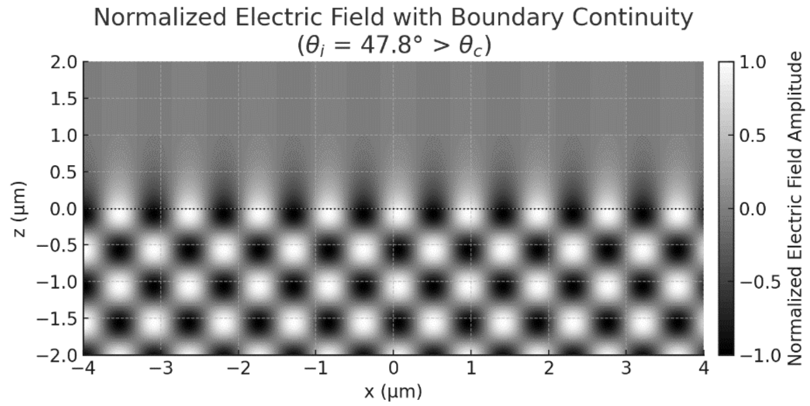

Figure. Distribution of the electric field at the interface (z = 0) under total internal reflection. The dotted line represents the boundary.

2. Waveguides

1) Basic Concepts

We learned in the previous section that when a beam inside a high-index medium meets a single interface with a lower-index medium at an angle above the critical value, total internal reflection keeps energy from leaking out. Now imagine placing a second, parallel interface below the first at just the right spacing. If that spacing lets each round-trip reflection interfere constructively with the next, forming a standing-wave envelope in the transverse (z) direction while the whole pattern slides forward along x at the phase velocity $\mathrm{v}_{!\mathrm{p}} = \omega/k_x$. The result is a planar dielectric waveguide: light is trapped vertically by total internal reflection yet propagates horizontally with very little loss.

The figure below visualizes one such guided mode. The grayscale map shows the electric field amplitude in a 1.69 µm-thick glass core ($n_1 = 1.5$) sandwiched between air claddings ($n_2 = 1.0$). Dotted lines mark the two core–cladding boundaries. Inside the slab bright–dark lobes can be seen, indicating that full cosine–like oscillation fits across the thickness; outside, the field decays evanescently, confirming that energy remains confined while the wavefronts march to the right.

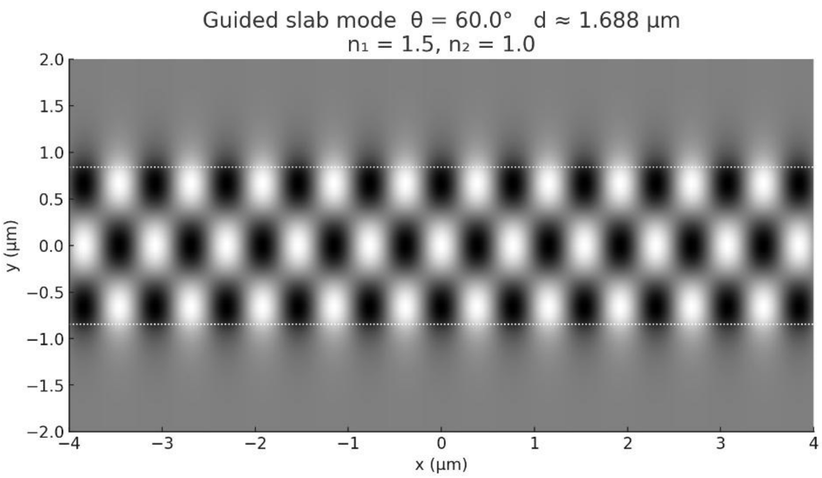

Figure. Electric-field map of a guided mode in a planar dielectric slab.

The dotted lines represent the boundaries.

Analyzing a dielectric slab waveguide is complicated by the phase shifts that accumulate at each instance of total internal reflection. This topic will be explored in more detail in your homework. For introductory purposes, it is instructive to consider the metallic waveguide, whose walls are idealized as perfect conductors. Such boundaries produce total reflection without the additional phase-shift complications, allowing the modal structure to be derived more straightforwardly.
It is important to recognize, however, that perfect conductors do not exist in practice; real metals exhibit finite conductivity. Consequently, metallic waveguides typically suffer higher attenuation than dielectric waveguides that rely on total internal reflection—most notably optical fibers—where propagation losses can be orders of magnitude lower.

2) Rectangular Metallic Waveguides

① $\mathbf{TE_{10}}$ Mode

We idealize the waveguide walls as perfect electric conductors (PECs). At a PEC interface the tangential electric field and the normal magnetic flux density must vanish:

$$\boxed{\vec{E}_{\parallel} = 0, \qquad \vec{H}_{\perp} = 0}.$$

To follow the usual waveguide convention, we now take the z-axis as the guiding (propagation) direction and the x-axis as the principal transverse direction, effectively rotating the coordinate system used in the previous section by 90 °. The updated geometry is sketched in the following figure, where the waveguide walls are at $x = 0,\ x = a$ and all fields are assumed to vary as $e^{\,j(k_z z - \omega t)}$.

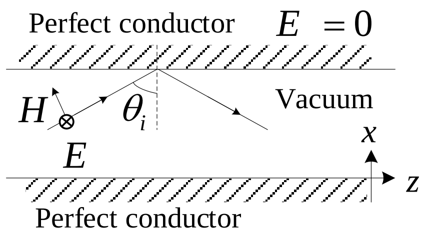

Figure. A metallic slab waveguide showing the field configurations.

The figure illustrates the field behavior for an S-polarized wave. At each reflection from the PEC, the phase of the electric field undergoes a π shift. This ensures that the incident and reflected electric fields cancel exactly at the conducting boundary, satisfying the boundary condition

$$\vec{E}_{\parallel} = 0.$$

This cancellation occurs at both the top and bottom walls of the waveguide (e.g., at $x = 0$ and $x = a$), enforcing nodes in the electric field at these positions. As a result, the only permissible electric field distributions along the transverse (x) direction are standing waves of the form:

$$\vec{E}(x) \propto \sin\left(\frac{m\pi x}{a}\right), \quad m = 1,2,3,$$

As we will learn soon, these are the transverse field profiles of the allowed TE_{m0} modes. The sinusoidal variation ensures that the boundary condition is satisfied at the conducting walls, and that the field is self-consistent after multiple reflections inside the waveguide.

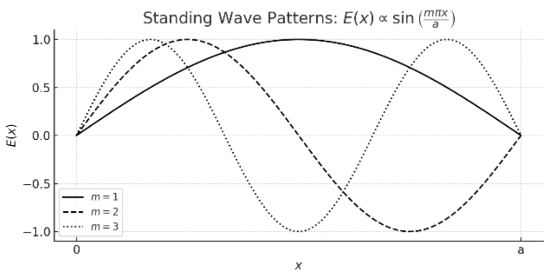

Figure. Transverse electric field pattern corresponding to m = 1, 2, 3.

It appears that placing two additional metal plates parallel to the x-z plane, i.e., at fixed y-coordinates and separated by an appropriate distance, does not violate the boundary conditions. Since the electric field is oriented perpendicular to these planes, the tangential component $\vec{E}_{\parallel}$ vanishes at the conducting surfaces, thereby satisfying the boundary condition.

Now a rectangular metallic waveguide is formed by enclosing the core region with perfectly conducting walls on all sides. The electric field is transversely polarized and confined within the rectangular cross-section, forming standing wave patterns that satisfy the boundary condition at the conducting surfaces. These transverse field distributions are sustained by constructive interference under the waveguide's geometric constraints. As a result, the electromagnetic wave is effectively trapped in the transverse direction and can propagate along the longitudinal (z) axis without radiative loss. The modes we have established so far are all denoted as TE_{m0}, where "TE" indicates that the electric field has no component along the propagation direction ($E_z = 0$), and the indices m and 0 specify the number of half-wavelength variations of the field in the x- and y-directions, respectively. The electric field lines of TE_{10} mode at a fix instant are illustrated in the figure below.

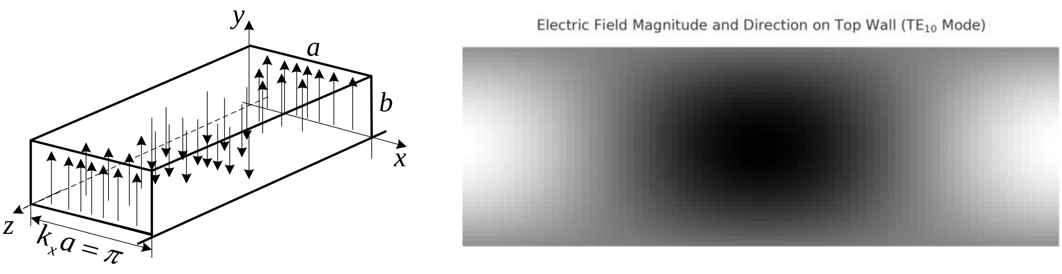

Figure. Electric fields lines of the TE_{10} mode and field distribution on the top wall.

The electric field of the TE_{10} mode is:

$$\tilde{\vec{E}}(x, y, z, t) = \hat{y}\tilde{E}_0 \sin\left(\frac{\pi x}{a}\right) e^{j(\beta z - \omega t)},$$

where $\beta = k_z$ is the propagation constant.

To compute the magnetic field and surface current distributions for the TE_{10} mode in a rectangular waveguide, we begin with the above expression for the electric field. Applying Faraday's law in its time-harmonic form, $\nabla \times \tilde{\vec{E}} = -j\omega\mu\tilde{\vec{H}}$, we obtain the magnetic field components H_x and H_z as spatial derivatives of E_y. These components define a circulating magnetic field pattern in the x-z plane. The surface current density on the conducting walls is then determined using the boundary condition $\tilde{\vec{J}}_s = \hat{n} \times \tilde{\vec{H}}$, where $\hat{n}$ is the outward normal to the surface. This yields current distributions that are consistent with the magnetic field circulation and satisfy the PEC boundary conditions. The directions of the magnetic field and surface current density on the top surface of the rectangular waveguide are illustrated in the figure below using arrows. The magnetic field forms closed loops that circulate around regions of strongest electric field variation, while the surface current flows tangentially to the walls, perpendicular to the magnetic field at the boundary.

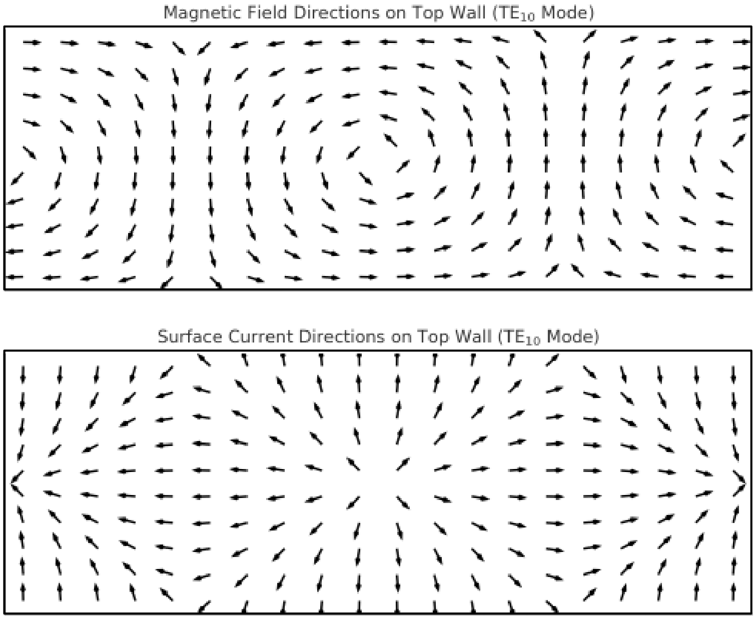

Figure. Magnetic field and surface current directions on the waveguide's top wall (TE_{10} mode).

② **Other TE Modes**

TE_{01} mode:

The roles of *x* and *y* are interchanged:

$$\tilde{\vec{E}}(x,y,z,t)=\hat{x}\tilde{E}_0\sin\left(\frac{\pi y}{b}\right)e^{j(\beta z-\omega t)}.$$

Note, in a rectangular waveguide, a and b represent the width and height of the waveguide cross-section, respectively, where $a > b$ by design convention.

Higher-order TE Modes:

Higher-order modes, TE_{mn} with $m \geq 1,\, n \geq 1$, are obtained by superposition of the standing-wave factors $\sin(k_x x)$ and $\sin(k_y y)$ with the transverse wavenumbers:

$$k_x=\frac{m\pi}{a},\quad k_y=\frac{n\pi}{b},\quad m,n=1,2,3$$

For all cases (including TE_{10} and TE_{01} modes) the total wavenumber satisfies

$$k^2=k_x^2+k_y^2+k_z^2=\left(\frac{\omega}{c}\right)^2.$$

③ **TM Modes:**

Transverse magnetic (TM) modes in a rectangular waveguide are characterized by having no magnetic field component along the direction of propagation, i.e., $H_z=0$, while the electric field has a nonzero longitudinal component $E_z \neq 0$. The field distributions for TM_{mn} modes are determined by solving Maxwell's equations with the boundary condition that the tangential component of the electric field must vanish at the conducting walls. This leads to standing-wave solutions for E_z of the form

$$\sin\left(\frac{m\pi x}{a}\right)\sin\left(\frac{n\pi y}{b}\right),$$

where $m, n \geq 1$. Unlike TE modes, there is no TM_{m0} or TM_{0n} mode since $E_z = 0$ on the walls implies both *m* and *n* must be nonzero. TM modes have higher cutoff frequencies than TE_{10}, and are generally not used unless specific mode control or higher-order propagation is desired.

④ Cut Off of Propagation Modes

We begin with the dispersion relation in a rectangular waveguide:

$$k_x^2 + k_y^2 + k_z^2 = k^2 = \left(\frac{\omega}{c}\right)^2,$$

with the resonance conditions:

$$k_x = \frac{m\pi}{a}, \quad k_y = \frac{n\pi}{b}.$$

Substituting into the dispersion relation gives:

$$k_z = \beta = \sqrt{\left(\frac{\omega}{c}\right)^2 - \left(\frac{m\pi}{a}\right)^2 - \left(\frac{n\pi}{b}\right)^2}.$$

Clearly, for a given mode index pair (*m*, *n*), if the frequency ω is too low, the square root becomes imaginary. This means k_z is purely imaginary, and the wave is evanescent—it cannot propagate and is said to be cut off. For that mode, the cut off (angular) frequency is:

$$\omega_c = c\sqrt{\left(\frac{m\pi}{a}\right)^2 + \left(\frac{n\pi}{b}\right)^2}.$$

⑤ Phase and Group Velocity in Waveguides

The phase velocity of the wave is:

$$v_p = \frac{\omega}{\beta} > \frac{\omega}{k} = c,$$

so the guided phase velocity always exceeds the speed of light *c*.

The group velocity, defined as:

$$v_g = \frac{d\omega}{d\beta} = \left(\frac{dk_z}{d\omega}\right)^{-1},$$

can be derived as:

$$v_g = c \cdot \sqrt{1 - \left(\frac{\omega_c}{\omega}\right)^2} < c.$$

Thus, while phase velocity exceeds *c*, group velocity—which carries energy and information—remains below *c*.

Dispersion in rectangular metallic waveguides arises because different frequency components of a signal propagate at different phase and group velocities. Unlike in non-dispersive media where phase velocity is constant, the phase velocity in a waveguide depends on frequency and is always greater than the speed of light, $v_p = \omega/k_z > c$. Conversely, the group velocity, which represents the speed of energy or information transfer, is always less than *c* and given by $v_g = d\omega/dk_z$. As frequency approaches the cutoff frequency of a given mode, the group velocity drops to zero and the wave becomes evanescent. This frequency-dependent propagation leads to modal dispersion, where different frequency components of a signal spread out over time, potentially distorting broadband pulses. To minimize dispersion and maintain signal integrity, waveguides are often operated in the single-mode regime, where only the fundamental TE_{10} mode is allowed to propagate.

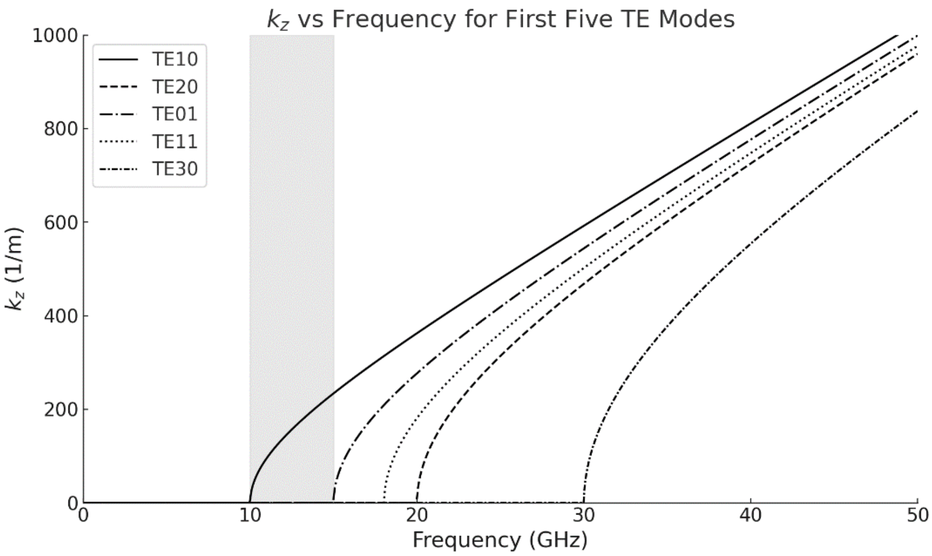

Figure. Dispersion curves of a rectangular waveguide with *a* = 1.5 cm, *b* = 1 cm. The curves of the five lowest TE modes are plotted. The shaded area represents the frequency range for single-mode operation, where only the TE_{10} mode can propagate.

The figure above shows the longitudinal wave number k_z (i.e., the propagation constant β) as a function of frequency for the first five TE modes in a rectangular metallic waveguide with dimensions *a* = 1.5 cm and *b* = 1 cm. Each curve begins at its respective cutoff frequency, where

$k_z = 0$, and increases with frequency, indicating the onset and growth of wave propagation for that mode. Below the cutoff, k_z is imaginary, corresponding to evanescent (non-propagating) behavior. The shaded region highlights the frequency range between 10 GHz and the cutoff frequency of the second mode (TE_{01}), where only the TE_{10} mode can propagate—this is the <u>single-mode region</u>. Within this band, signal transmission avoids modal dispersion, making it ideal for high-fidelity communication. Above this region, higher-order modes begin to propagate, each with its own frequency-dependent k_z, leading to multimode dispersion. The figure clearly illustrates how cutoff and dispersion are intrinsic to waveguide behavior and emphasizes the practical importance of operating within the single-mode regime.